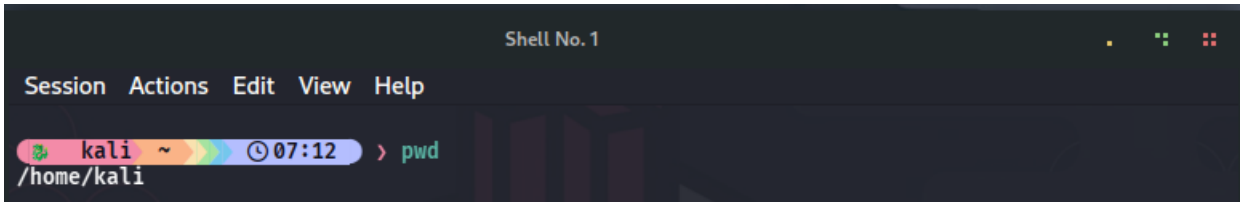
EX: 2	Practice the Basic Commands in Kali Linux
DATE:	

Objective

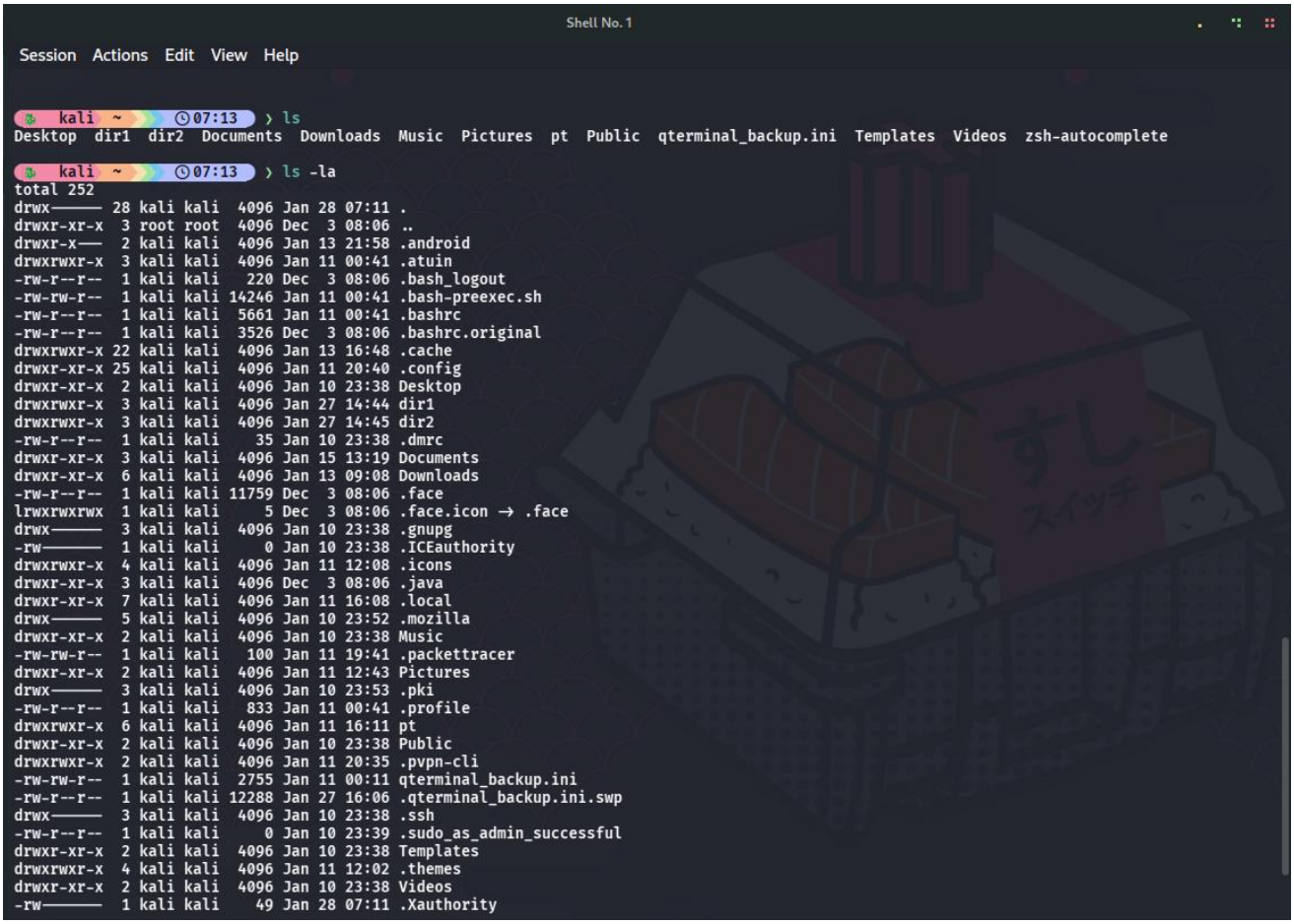
To understand and practice essential Linux commands used in Kali Linux for cybersecurity tasks.

Steps

1. Boot the system using Kali Linux (Virtual Machine or Live Boot).
2. Open the **Terminal**.
3. Practice file and directory commands:
    - **pwd** – Display present working directory



    - **ls, ls -la** – List files and directories



- **cd** – Change directory

```
Shell No. 1
Session  Actions  Edit  View  Help
kali ~ 07:15 ❯ ls
Desktop  dir1  dir2  Documents  Downloads  Music  Pictures  pt  Public  qterminal_backup.ini  Templates  Videos  zsh-autocomplete
kali ~ 07:15 ❯ cd Desktop
kali ~/Desktop 07:15 ❯ ls
kali ~/Desktop 07:15 ❯
```

4. Practice file operations:
   - **touch** filename

```
kali ~/ 07:16 ❯ touch file.txt
kali ~/ 07:17 ❯ ls
file.txt  hack1-01.cap  learn  Onion  proton  wifi.hc22000
```

   - **cat** filename

```
Shell No. 1
Session  Actions  Edit  View  Help
kali ~/ 07:18 in 9s388ms ❯ cat file.txt
Hello world!
```

   - **cp source** destination

```
kali ~/ 07:19 ❯ cat file2.txt
kali ~/ 07:19 ❯ cp file file2
cp: cannot stat 'file': No such file or directory
kali ~/ 07:19 ❯ cp file.txt file2.txt
kali ~/ 07:19 ❯ cat file2.txt
Hello world!
```

   - **mv** oldname newname

```
kali ~/ 07:20 ❯ mv file.txt file2.txt
kali ~/ 07:20 ❯ cat file.txt
cat: file.txt: No such file or directory
```

   - **rm** filename

```
kali ~/ 07:21 ❯ ls
file2.txt  hack1-01.cap  learn  Onion  proton  wifi.hc22000
kali ~/ 07:21 ❯ rm file2.txt
kali ~/ 07:22 ❯ ls
hack1-01.cap  learn  Onion  proton  wifi.hc22000
```

5. **Practice system commands:**

   - **ifconfig / ip a**

```
kali ~/ 07:22 ❯ ifconfig
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1500
        inet 192.168.209.132  netmask 255.255.255.0  broadcast 192.168.209.255
        inet6 fe80::4f1c:9edf:98c4:c967  prefixlen 64  scopeid 0x20<link>
        ether 00:0c:29:08:60:27  txqueuelen 1000  (Ethernet)
        RX packets 93  bytes 15734 (15.3 KiB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 115  bytes 10675 (10.4 KiB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING>  mtu 65536
        inet 127.0.0.1  netmask 255.0.0.0
        inet6 ::1  prefixlen 128  scopeid 0x10<host>
        loop  txqueuelen 1000  (Local Loopback)
        RX packets 8  bytes 480 (480.0 B)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 8  bytes 480 (480.0 B)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0
```

   - **uname -a**

```
kali ~/ 07:26 ❯ uname -a
Linux kali 6.17.10+kali-amd64 #1 SMP PREEMPT_DYNAMIC Kali 6.17.10-1kali1 (2025-12-08) x86_64 GNU/Linu
```

   - **whoami**

```
kali ~/ 07:27 ❯ whoami
kali
```

6. **Practice permission commands:**

   - **Chmod**

```
kali ~/ 07:27 ❯ touch file2.txt
kali ~/ 07:27 ❯ chmod 777 file2.txt
kali ~/ 07:28 ❯ ls -l file2.txt
-rwxrwxrwx 1 kali kali 0 Jan 28 07:27 file2.txt
```

   - **Chown**

```
┌──(kali㉿kali)-[~]
└─$ sudo chown root ex
[sudo] password for kali:
Sorry, try again.
[sudo] password for kali:

┌──(kali㉿kali)-[~]
└─$ ls -l ex
-rwxrwxrwx 1 root kali 0 Jan 27 05:28 ex

┌──(kali㉿kali)-[~]
└─$
```

7. Practice networking commands:

- **Ping** any ip

```
┌──(root㉿kali)-[/home/kali/Documents]
└─# ping skcet.ac.in
PING skcet.ac.in (88.222.243.2) 56(84) bytes of data.
64 bytes from 88-222-243-2.init.lt (88.222.243.2): icmp_seq=1 ttl=128 time=169 ms
64 bytes from 88-222-243-2.init.lt (88.222.243.2): icmp_seq=2 ttl=128 time=173 ms
64 bytes from 88-222-243-2.init.lt (88.222.243.2): icmp_seq=3 ttl=128 time=175 ms
64 bytes from 88-222-243-2.init.lt (88.222.243.2): icmp_seq=4 ttl=128 time=178 ms
64 bytes from 88-222-243-2.init.lt (88.222.243.2): icmp_seq=5 ttl=128 time=177 ms
64 bytes from 88-222-243-2.init.lt (88.222.243.2): icmp_seq=6 ttl=128 time=172 ms
64 bytes from 88-222-243-2.init.lt (88.222.243.2): icmp_seq=7 ttl=128 time=204 ms
^C
--- skcet.ac.in ping statistics ---
7 packets transmitted, 7 received, 0% packet loss, time 6006ms
rtt min/avg/max/mdev = 169.106/178.442/204.461/11.014 ms
```

- **Netstat** any ip

```
kali ~/Ⓑ  07:31  in 2m11s211ms ❯ netstat 8.8.8.8
Active Internet connections (w/o servers)
Proto Recv-Q Send-Q Local Address           Foreign Address         State
udp        0      0 192.168.209.132:bootpc  192.168.209.254:bootps  ESTABLISHED
Active UNIX domain sockets (w/o servers)
Proto RefCnt Flags       Type       State         I-Node   Path
unix  3      [ ]         STREAM     CONNECTED     19543    /run/dbus/system_bus_socket
unix  3      [ ]         STREAM     CONNECTED     17666    /run/user/1000/at-spi/bus_0
unix  3      [ ]         STREAM     CONNECTED     14151    /run/dbus/system_bus_socket
unix  3      [ ]         STREAM     CONNECTED     17814
unix  3      [ ]         STREAM     CONNECTED     20511    /run/user/1000/at-spi/bus_0
unix  3      [ ]         STREAM     CONNECTED     15097    @/tmp/.ICE-unix/1303
unix  10     [ ]         DGRAM      CONNECTED     11554    /run/systemd/journal/dev-log
unix  3      [ ]         STREAM     CONNECTED     20587
unix  3      [ ]         STREAM     CONNECTED     19529    @/tmp/.ICE-unix/1303
unix  3      [ ]         STREAM     CONNECTED     15812
unix  3      [ ]         STREAM     CONNECTED     3971
unix  11     [ ]         DGRAM      CONNECTED     11555    /run/systemd/journal/socket
unix  3      [ ]         STREAM     CONNECTED     13277
unix  3      [ ]         STREAM     CONNECTED     8900     /run/dbus/system_bus_socket
unix  3      [ ]         DGRAM      CONNECTED     13098
unix  3      [ ]         DGRAM      CONNECTED     11533
unix  3      [ ]         STREAM     CONNECTED     14210
unix  3      [ ]         STREAM     CONNECTED     15541
unix  3      [ ]         STREAM     CONNECTED     16046
unix  3      [ ]         STREAM     CONNECTED     15081
unix  3      [ ]         STREAM     CONNECTED     16040
unix  3      [ ]         STREAM     CONNECTED     21531    /run/dbus/system_bus_socket
unix  3      [ ]         STREAM     CONNECTED     13304
unix  3      [ ]         STREAM     CONNECTED     13251    /run/user/1000/bus
unix  3      [ ]         STREAM     CONNECTED     15849    /run/user/1000/pipewire-0-manager
unix  3      [ ]         STREAM     CONNECTED     18819    /run/user/1000/bus
unix  3      [ ]         STREAM     CONNECTED     17758
unix  3      [ ]         STREAM     CONNECTED     14208
unix  3      [ ]         STREAM     CONNECTED     19663
unix  3      [ ]         STREAM     CONNECTED     15654    @/tmp/.X11-unix/X0
unix  3      [ ]         STREAM     CONNECTED     8899
unix  2      [ ]         DGRAM      CONNECTED     12550
unix  3      [ ]         STREAM     CONNECTED     17674    /run/user/1000/bus
unix  3      [ ]         DGRAM      CONNECTED     15490
unix  3      [ ]         STREAM     CONNECTED     15885    /run/user/1000/at-spi/bus_0
unix  3      [ ]         STREAM     CONNECTED     15028    /run/dbus/system_bus_socket
unix  3      [ ]         STREAM     CONNECTED     14136    /run/user/1000/bus
unix  3      [ ]         STREAM     CONNECTED     18691    /run/user/1000/pipewire-0
unix  3      [ ]         STREAM     CONNECTED     15141
unix  3      [ ]         STREAM     CONNECTED     15104
```

- **traceroute** any ip

```
kali ~/Ⓑ  07:32 ❯ traceroute 8.8.8.8
traceroute to 8.8.8.8 (8.8.8.8), 30 hops max, 60 byte packets
 1  192.168.209.2 (192.168.209.2)  0.461 ms  0.390 ms  0.354 ms
 2  * * *
 3  * * *
 4  * * *
 5  * * *
 6  * * *
 7  * * *
 8  * * *
 9  * * *
10  * * *
11  * * *
12  * *^C
```

Result

Basic Kali Linux commands were successfully practiced and understood.